

COURSE TITLE : FOUNDATIONS IN SCIENCE & TECHNOLOGY
COURSE CODE : 1252
SEMESTER : 1
COURSE CATEGORY : F
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	METHOD OF SCIENCE	15
2	ECOLOGY AND ENVIRONMENT	15
3	DYNAMICS OF PARTICLES AND NEWTONS LAW OF MOTION	15
4	ATOMIC NUCLEUS	15

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand methods of science
	2	To understand nature of scientific knowledge
2	1	To understand eco systems and its importance
	2	To understand the concept of pollution and its effects
3	1	To understand measurement of concepts and laws
4	1	To understand the concept and force of atomic nucleus

SPECIFIC OUTCOMES

1.1.0 Understand the method of science concept

- 1.1.1 Explain the steps in scientific method i.e. observation, hypothesis, experiments and apparatus
- 1.1.2 Explain how laws, models and theories are developed
- 1.1.3 List some examples to illustrate the method of science – preparation of food by plants, invention of barometer, burning of candle and conservation of mass

1.2.0 Understand the nature of scientific knowledge

- 1.2.1 Explain the dynamic, never complete, repeatable, verifiable, open to change and self correcting nature of scientific knowledge

1.3.0 Understand scientific nature to problem solving

- 1.3.1 Explain use of scientific reasoning and logic to solve problems Outline the strategy of science
- 1.3.2 Describe how scientific achievements are used for better living and removal of ignorance

2.1.0 Understand Eco system and environment

- 2.1.1 Explain – ‘individuals’ , ‘photo synthesis’ and ‘respiration’
- 2.1.2 Define food chain and food web

2.2.0 Understand science as a way of thinking

2.2.1 Explain the many facets of science

2.2.2 Explain the demarcation of science-physics, mathematics, biology, medicine etc.

2.3.0 Know the various interaction in eco system and cycling of materials

2.3.1 Explain cycling of materials ,nitrogen cycle ,carbon cycle and water cycle

2.3.2 Explain ecology crisis

2.4.0 Understand pollution and its effect

2.4.1 Explain pollution –pollution from burning forests, photochemical smog, acid rain-effect of pollution on weather -green house effect , deflection of ozone layer

2.4.2 Narrate steps to control pollution

2.5.0 Understand impact of technology on environment

2.5.1 Explain in brief Bhopal disaster, Chernobyl disaster

2.5.2 Explain the effect of population increase on environment

3.1.0 Understand measurement cocepts and practice

3.1.1 Explain Measuring Units – Systems of Units

3.1.2 Explain Fundamental and derived Units –s.i.units

3.1.3 Explain Dimensions- Dimensions for different physical quantities

3.1.0 Understand dynamics of particles and newtons laws of motion

3.2.1 Define displacement, velocity, acceleration, uniform velocity, and uniform acceleration

3.2.2 State Equations of Motion

3.2.3 Solve simple problems on equation of motion

3.2.4 Explain Motion under gravity and 'g'.

3.3.0 Understand Newtons Laws and its application

3.3.1 State Newton's first law-inertia and force concept and momentum.

3.3.2 State Newton's second law-derive expression $f=mg$.

3.3.3 State Newton's third law-quote examples in daily life,.

3.3.4 Define conservation of momentum. , recoil of gun

4.1.0 Understand Atomic Nucleus

4.1.1 Explain Structure of Nucleus

4.1.2 Explain Nuclear Force

4.1.3 Define Radioactivity

4.1.4 Explain Nuclear Energy, Distinguish between Nuclear Fission and Nuclear Fussion

4.1.5 Explain Nuclear Holocaust

4.1.6 Explain Radiation Hazards

CONTENT DETAILS

- UNIT I Science as a human endeavour-achievements-history-institutions-aim-method-Various Operations-strategy-role in production.
- UNIT II Facets of science-demarcation of science-method-steps in scientific method-nature of Scientific knowledge-scientific nature to problem solving-scientific reasoning and -reflections of scientific achievements.
- UNIT III Eco system-cycling of materials, nitrogen cycle, Carbon cycle, water cycle-ecology crisis- pollution, Ozone layer-control of pollution-impact of technology on environment-Bhopal disaster, Chernobyl disaster.
- UNIT IV Development of science - technological development in India - technology in various industries-economic development and self reliance-technology policy-technology transfer - technology developments in engineering sector.

BOOKS RECOMMENDED

FST-I-Foundation course in Science and Technology-Block 1,2,4,7 & 8 (Total Rs.100/-) printed materials are available with the Joint Registrar, Material Distribution Division, IGNOU, Maidan Garhy, New Delhi-110 068.

1. R.V.G Menon , History of Science and Technology,Pearson
2. Arthur Beiser, Modern Physics, Tata McGraw